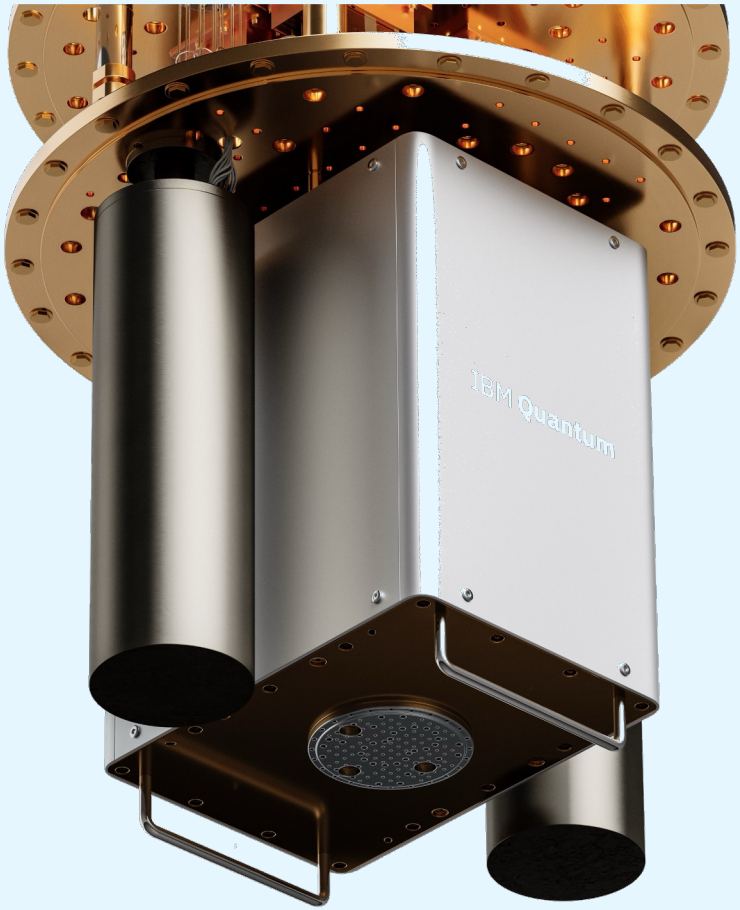
QAOA seminar: Post-processing



Kunal Kumar

Quantum Algorithm Engineer

https://www.linkedin.com/in/kunarukumaru/

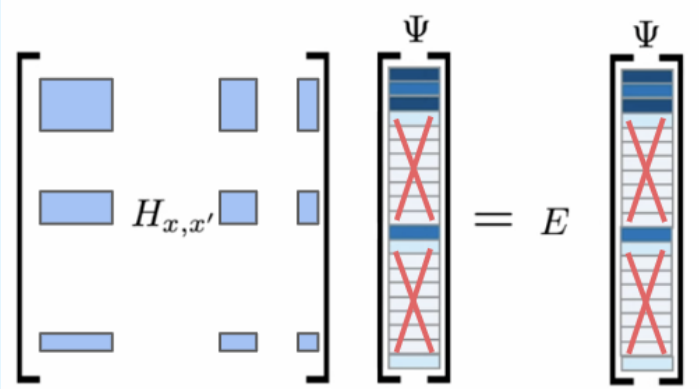
kunalkumar@ibm.com

23rd April, 2026

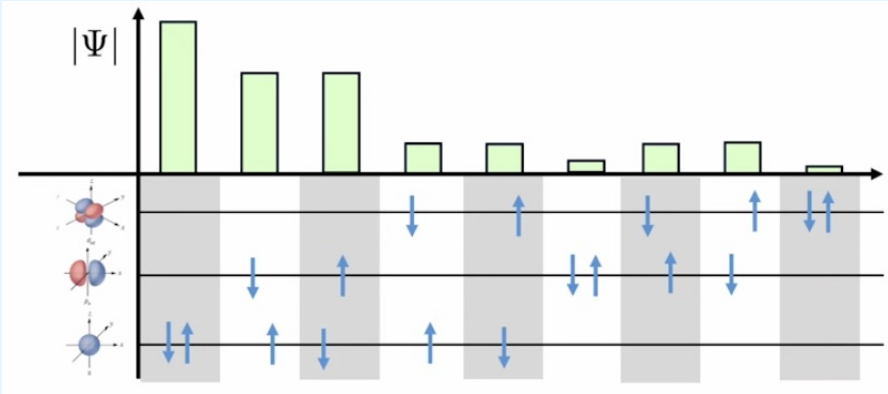
© 2026 IBM Corporation

Motivation

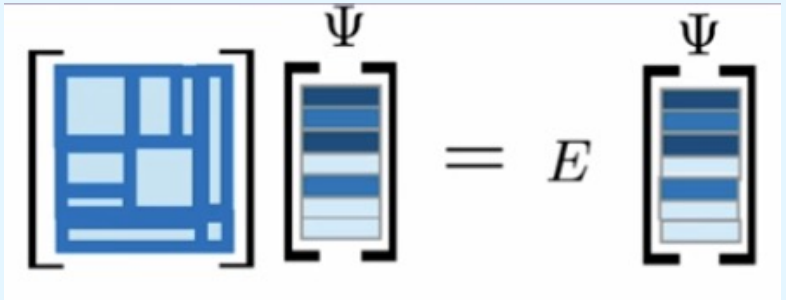
Sample-based Quantum Diagonalization (SQD)

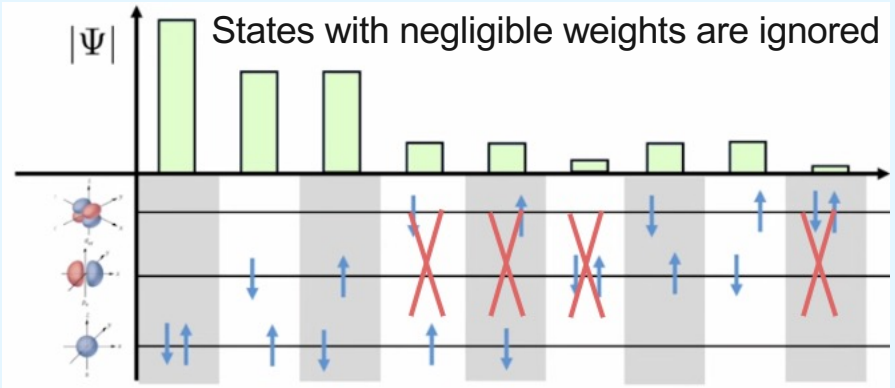


$H\,\Psi = E\Psi$



dimension is often too large for exact diagonalization





© 2026 IBM Corporation

https://arxiv.org/pdf/2405.05068

Greedy 1-flip local-search algorithm

```
        0 1 0 0
Step 0:
Row1: □ ■ □ □
Row2: □ □ ■ □
Row3: ■ □ □ ■

Step 1:
Row1: □ ■ □ □
Row2: □ □ ■ □
Row3: ■ ■ □ ■   Flip (Row3, Col2): □ → ■

Step 2:
Row1: □ ■ □ □
Row2: ■ □ ■ □   Flip (Row2, Col1): □ → ■
Row3: ■ ■ □ ■

Step 3:
Row1: □ ■ □ ■   Flip (Row1, Col4): □ → ■
Row2: ■ □ ■ □
Row3: ■ ■ □ ■
```

When you scan through all 12 bits and **no single flip** increases the cut, you stop

1. Input: bitstring x
2. Repeat:
   - For each index i:
     - Let $x^{(i)}$ be x with bit i flipped
     - Compute $\Delta = \mathrm{cut}(x^{(i)}) - \mathrm{cut}(x)$
   - If all $\Delta \leq 0$: stop
   - Else set $x \leftarrow x^{(i^*)}$ for the best/first improving i^*
3. Output: locally improved bitstring x

MaxCut

$$G = (V, E)$$

Constrain-based optimization

$$f(x) = x^T Q x + c^T x, \quad x \in \{0,1\}^n$$

© 2026 IBM Corporation

https://arxiv.org/pdf/1606.06833

IBM

Multi Qubit Correction

Max prob./Max eigenvalue (objective function)/ Hybrid

Seed: x

```
Row1: □ ■ □ □
Row2: □ □ ■ □
Row3: ■ □ □ ■
```

Reference: y

```
Row1: □ ■ ■ □
Row2: ■ □ ■ □
Row3: ■ □ □ □
```

D = { (1,3), (2,1), (3,4) }

```
x:  Row1: □ ■ □ □      y: Row1: □ ■ ■ □      diff at (Row1, Col3)
    Row2: □ □ ■ □         Row2: ■ □ ■ □      diff at (Row2, Col1)
    Row3: ■ □ □ ■         Row3: ■ □ □ □      diff at (Row3, Col4)
```

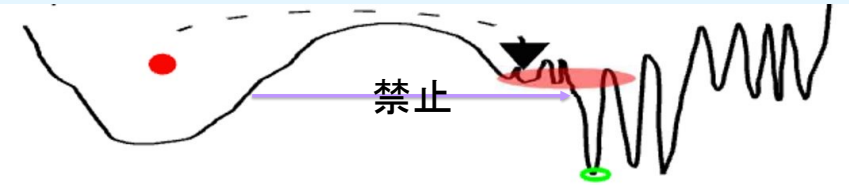
Group g1 = { (2,1), (1,3) } (connected)

Group g2 = { (3,4) } (isolated)

```
Before:
Row1: □ ■ □ □
Row2: □ □ ■ □
Row3: ■ □ □ ■

Flip g1 = {(Row2,Col1),(Row1,Col3)}:
Row1: □ ■ ■ □
Row2: ■ □ ■ □
Row3: ■ □ □ ■
```

Group g2 flip: flip (Row3,Col4)

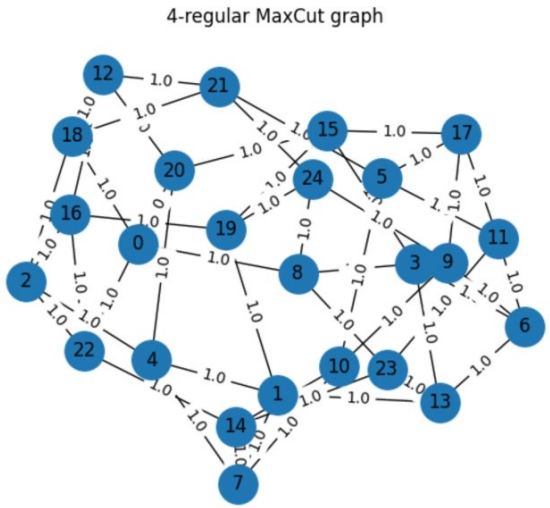
```
Row3: ■ □ □ ■  ->  ■ □ □ □
```

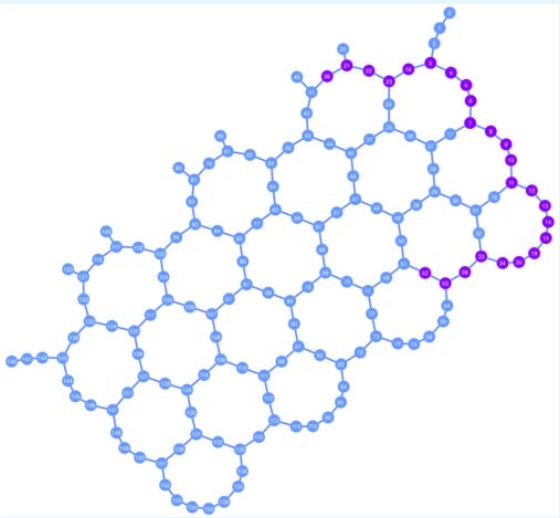



```
Input:
  - Top-M samples S = {(x_j, p_j)} sorted by probability
  - Graph G(V,E)
  - Max passes P
Output:
  - Postprocessed distribution over improved solutions

For each seed sample x in top-M:
  repeat P times:
    improved ← False
    For each reference sample y in top-M:
      D ← { i | x_i ≠ y_i }                        # differing indices
      Groups ← ConnectedComponents(G induced by D)
      For each group g in Groups:
        x' ← x with bits in g flipped
        if Cut(x') > Cut(x): candidate improvement
      If best improving group g* exists:
        x ← flip-group(x, g*)
        improved ← True
    If not improved: break

  x ← Greedy1FlipPolish(x)                          # final local refine
  Add probability mass p(seed) to bucket for x

Return normalized buckets as new distribution
```

© 2026 IBM Corporation

https://www.nature.com/articles/s41598-021-95482-w





Cost operator qubits: 25 Edges: 50 Classical best cut: 40.0
Classical best bit: [1, 0, 1, 0, 1, 1, 0, 1, 1, 1, 0, 1, 1, 1, 1, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0]

Backend: IBM Marrakesh [SAT] min_layers=9
Best path: [36, 21, 22, 23, 16, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 19, 35, 34, 33, 39, 53, 52]
Best path fidelity 0.9561951617850969 Num. evaluated paths 65877
(beta, gamma): (2.768861321807953, 5.750711976062672)

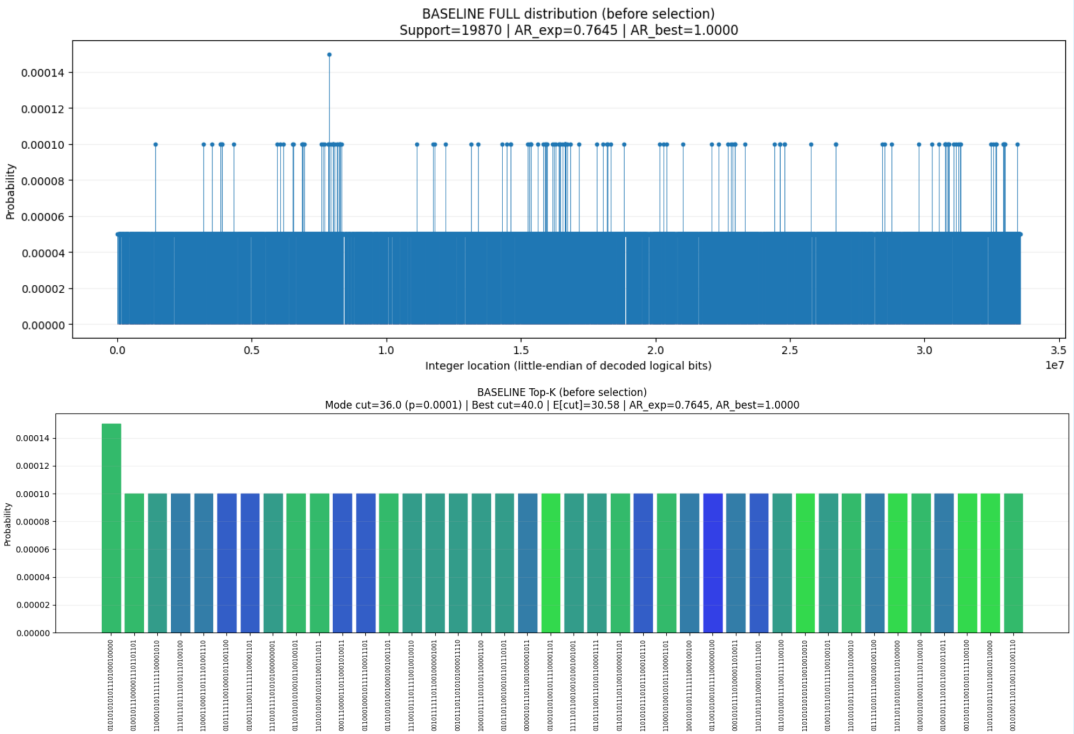
Mode bitstring: bitstring with the **highest probability**
Best bitstring: bitstring is the one with the **maximum cut value**
Expected cut: measures the **average solution quality**
Approximation Ratio (AR): $\frac{QAOA\ objective\ value}{optimal\ objective\ value}$

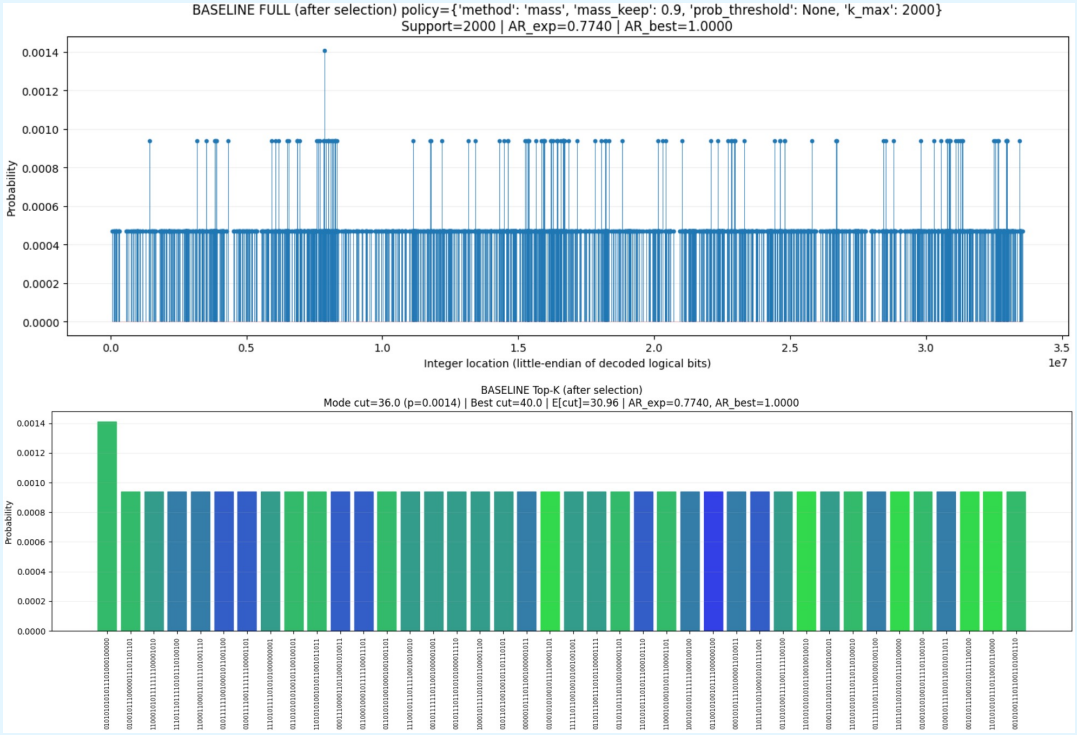
	variant	n	classical_opt	mode_bitstring	mode_prob	mode_cut
0	baseline	25	40.0	0101010101011101000100000	0.000150	36.0
1	baseline_selected	25	40.0	0101010101011101000100000	0.001408	36.0

	best_bitstring	best_prob	best_cut	expected_cut	ar_best	ar_expected	num_support
	0010101110010001101001101	0.000100	40.0	30.580100	1.0	0.764502	19870
	0010101110010001101001101	0.000939	40.0	30.960563	1.0	0.774014	2000

© 2026 IBM Corporation

Probability Distribution





© 2026 IBM Corporation

Post-processing

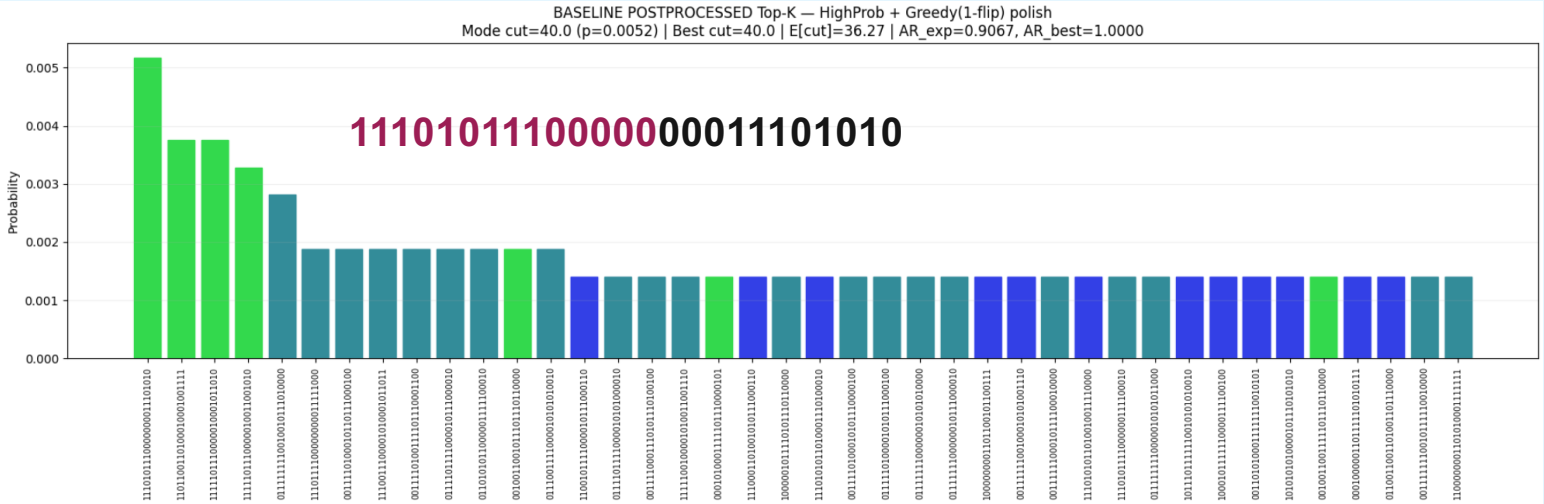
Variant	Post-processing method	Reference paper	Expected cut	Best cut	AR (expected)	AR (best)	Support
baseline	High-prob + Greedy (1-flip) polish	Chancellor (2017)	36.268545	40.0	0.906714	1.0	1798
baseline	MQC-style multi-qubit correction + polish	Ayanzadeh et al. (2021)	40.0	40.0	1.0	1.0	11

2000 seeds → multi-qubit group corrections → 11 distinct corrected solutions

Support = the number of **distinct bitstrings** that remain in the distribution *after post-processing*.

© 2026 IBM Corporation

Probability Distribution

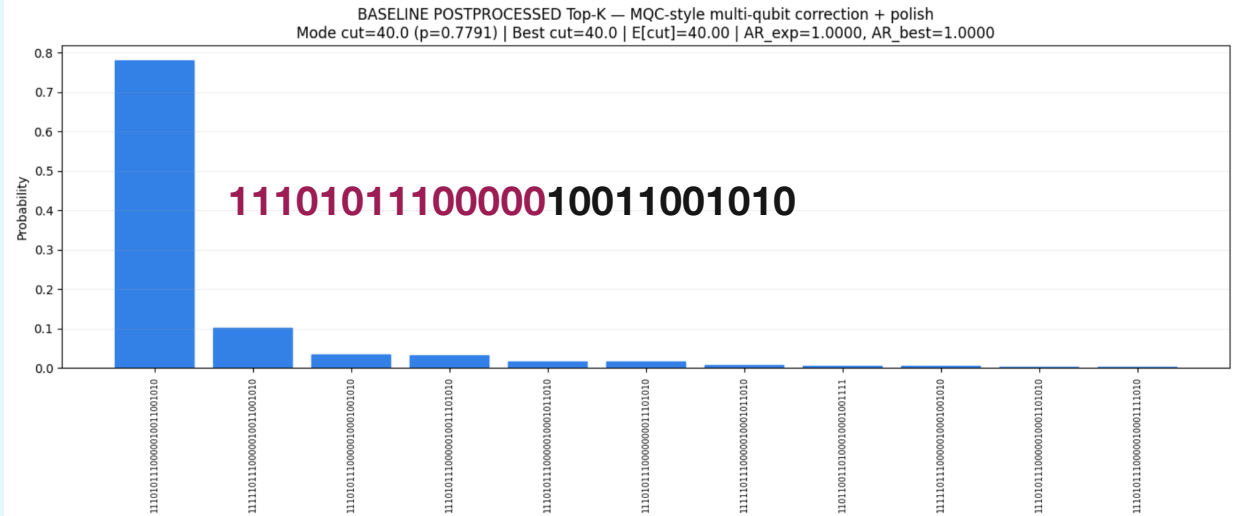


pos **15** **20**

A = 11101011100000**1**0011**0**01010

B = 11101011100000**0**0011**1**01010

Same local optimum



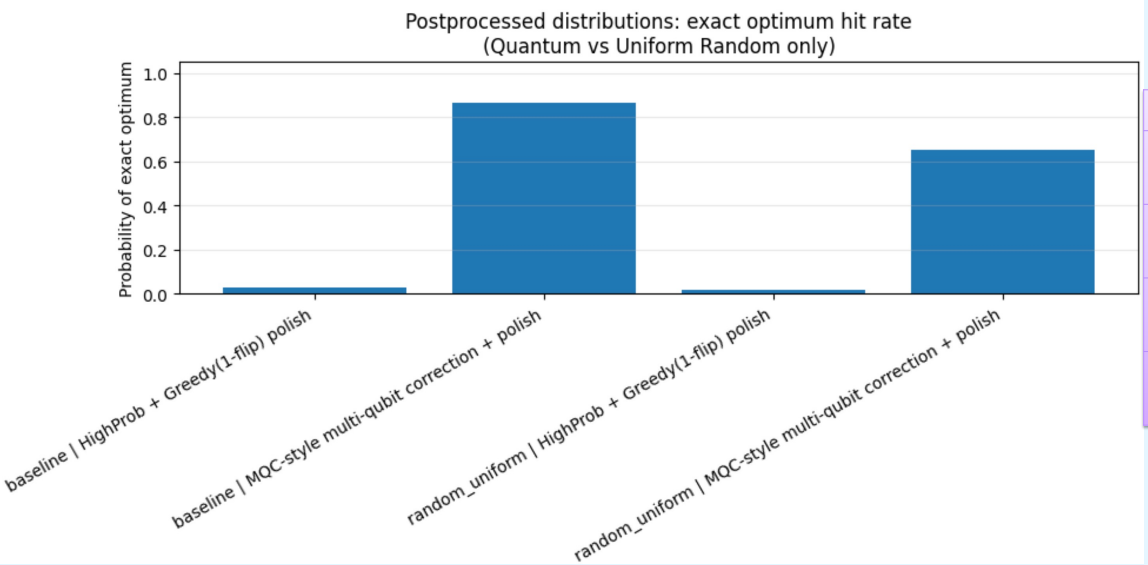
<u>Critical Hyperparameters to tune:</u>

Num of shots: must scale with problem size because the output distribution becomes **exponentially flatter** as the number of qubits increases

Selection criteria for the reference probabilities: Probability-based (small n) or Objective-based (large n), Reference pool size (20<n≤40: 200 – 500, 40<n≤60: 500 – 2000)

© 2026 IBM Corporation

Conclusion



Variant	P(cut>=39)	P(cut>=40)	P(optimum)
baseline | HighProb + Greedy(1-flip) polish	0.026066	0.026066	0.026066
baseline | MQC-style multi-qubit correction + polish	0.862010	0.862010	0.862010
random_uniform | HighProb + Greedy(1-flip) polish	0.016484	0.016484	0.016484
random_uniform | MQC-style multi-qubit correction + polish	0.650000	0.650000	0.650000

Thank you very much for listening

© 2026 IBM Corporation